

Distributional Semantic Models

Part 4: A linguistic perspective on DSMs

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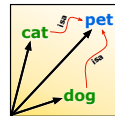
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<http://wordspace.collocations.de/doku.php/course:start>

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Does evaluation tell us everything?

A cautionary tale

Beyond NLP: Linguistic evaluation

Polysemy

Compositionality

Other linguistic aspects

The FAST task

Free association norms

FAST: Data set and tasks

FAST: Experiments

Hands-on exercises

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The CogALex-V Shared Task

Santus *et al.* (2016)

- ▶ Aim: better linguistic understanding of DS from identification of specific **semantic relations**
- ▶ Data: 747 target words with approx. 10 candidate relata each
 - ▶ training set: 318 targets, 3054 word pairs
 - ▶ test set: 429 targets, 4260 word pairs
- ▶ Subtask 1: related *vs.* unrelated word pairs
 - ▶ unrelated pairs are random fillers
 - ▶ relatively easy: $F_1 = 79.0\%$ (best system)
- ▶ Subtask 2: distinguish between semantic relations
 - ▶ SYN: w_2 can be used with same meaning as w_1
 - ▶ ANT: w_2 can be used as the opposite of w_1
 - ▶ HYPER: w_1 is a kind of w_2
 - ▶ PART_OF: w_1 is a part of w_2
 - ▶ RANDOM: no relation (random word + manual check)
 - ▶ relatively hard: $F_1 = 44.5\%$ (best system: **deep learning**)

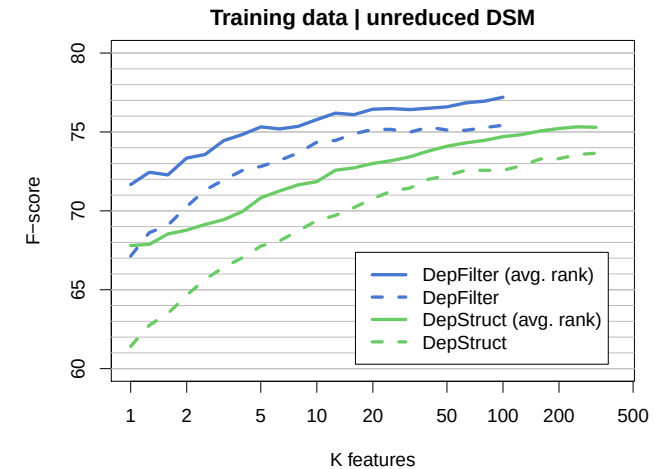
Mach 5 at CogALex 2016

Evert (2016)

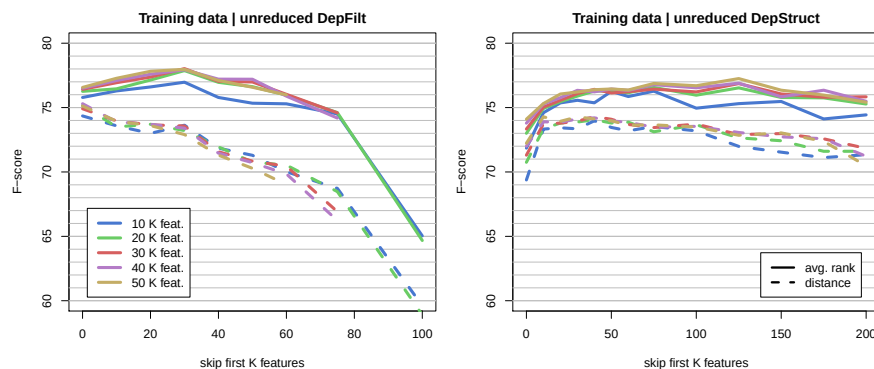
- ▶ Mach 5 participated in the CogALex-V Shared Task as a traditional “count” (non-neural) DSM
 - ▶ 10-billion-word Web corpus (Schäfer & Bildhauer 2012)
 - ▶ syntactic dependencies from C&C parser (Curran *et al.* 2007)
 - ▶ 26.5k target words, up to 300k feature dimensions
 - ▶ other parameters set according to Lapesa & Evert (2014)
- ▶ Parameter optimization on training data (subtask 1)
- ▶ Machine learning on optimized representations (subtask 2)
 - ▶ learns relevance weights for 600 latent SVD dimensions
 - ▶ best results from combination of different SVD spaces

Try it yourself: <http://www.collocations.de/data/#mach5>

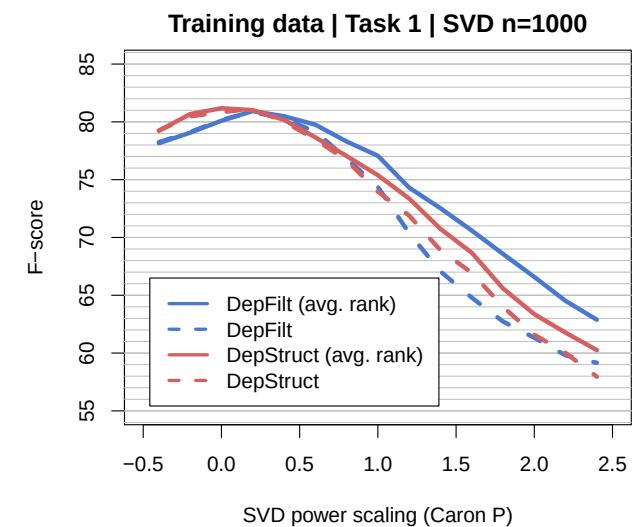
Mach 5: Parameter optimization



Mach 5: Parameter optimization



Mach 5: Parameter optimization



Mach 5: Are we doing well?

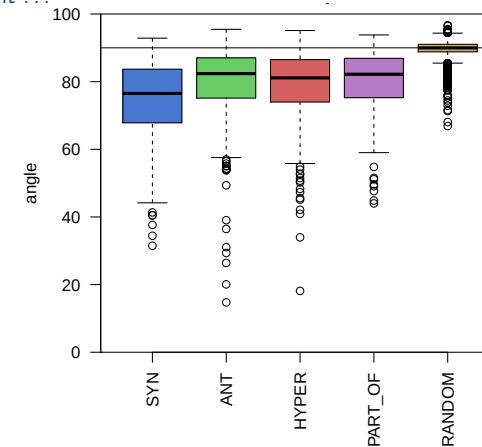
$F_1 = 77.88\%$ for related vs. unrelated (best: 79.0%)

However ...

- ▶ Parameter optimization yields surprising result:
best model uses < 50k features with relatively low frequency
- ▶ Nearest neighbours are unsatisfactory, e.g. for *play*:
playing (54.1°), *star* (62.8°), *reunite* (62.9°), *co-star* (64.3°),
reprise (64.4°), *player* (66.7°), *score* (68.5°), *audition* (69.2°),
sing (69.4°), *actor* (69.5), *understudy* (69.6), *game* (70.3), ...
- ▶ Why is Mach 5 still doing so well in the task, then?

Mach 5: What is going wrong?

A disturbing result ...



- DSM has learned to recognize random word pairs (at 90°)!
- We need better data sets with **high-quality distractors**!

Problems of standard tasks & data sets

Problems with semantic interpretation of DSMs don't only stem from evaluation methodology ...

... **data sets can be problematic as well!**

Two major problems:

- ▶ DSMs may exploit contingent properties of the task
 - ▶ **random fillers** as distractors ("controls")
 - ➔ recognize random word pairs rather than semantic relations
 - ▶ choice of clearly separated categories and prototypical exemplars in noun clustering task (ESSLI 2008)
 - ➔ much harder to identify categories in general word list
 - ▶ typical superordinate-level words in hypernym detection task
 - ➔ recognize "typical hypernym" in a multiple-choice setting
- ▶ Data set size too small
 - ▶ e.g. 97.5% accuracy on 80 TOEFL items ➔ over-fitting

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DSM similarity & linguistic theory

Numerical results from evaluation tasks can be misleading ...

📖 **perhaps a linguistic perspective gives us better insights?**

1. Polysemy
 - ▶ a textbook challenge, will look at most intuitive solution
 - 📖 hands-on: [hands_on_part4_schuetze1998.R](#)
2. Compositionality
 - ▶ above and below the word level
 - 📖 hands-on: derivational morphology (Lazaridou *et al.* 2013)
3. Not all meaning is distributional
 - ▶ e.g. function words, proper names
4. Distributional vectors as collocational profiles

Great overview paper:

[Distributional Semantics and Linguistic Theory](#) (Boleda 2020)

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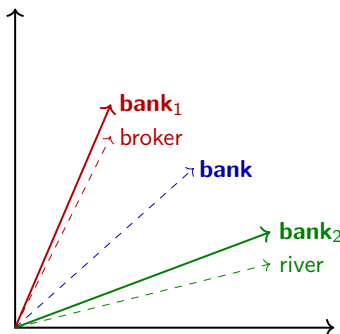
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Polysemy in DSMs

- ▶ **Problem:** DSM vectors conflate contexts from different senses
 - ▶ collocates of “bank”: *money, river, account, swim, ...*
 - ▶ conflated DSM vector either fits neither meaning well or is dominated by most frequent sense



Polysemy in DSMs

Observation: DSM vectors conflate contexts from word senses

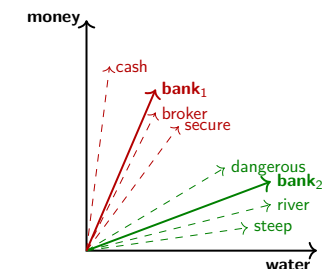
- ▶ **Solution:** build a representation for each instance of the word we want to disambiguate (Schütze 1998)

📖 sentence vectors

Target: bank

bank₁: *The broker went to the bank to secure his cash*

bank₂: *The river bank was steep and dangerous*



Application: word sense disambiguation

... can you think about another situation in which we may need it?

Context vectors: can we do them in **wordspace**?

Yes :-)

```
library(wordspace)
# S1: "Cats and dogs need their time"
s1 <- "cat and dog need their time"
# S2: "Time is the cause not the effect"
s2 <- "time is the cause not the effect"
# ingredients: vectors for individual words
>TT <- DSM_TermTermMatrix
>TT
      breed tail feed kill important explain likely
cat      84   17   8   38         0         2     0
dog     579   14  32   63         1         2     2
animal   45   11  86  136        13         5     4
time     19    8  29  134        94        44    100
reason    1    0   1   18        71       140     39
cause     0    1   0    3        55        35     51
effect    0    1   1    6        62        37     14
```

Context vectors: can we do the, in **wordspace**?

Yes :D

"cats and dogs need their time"

```
> context.vectors(TT, s1)
      breed tail feed      kill important explain likely
1 227.3333   13   23 78.33333 31.66667      16      34
# context.vectors() is taking the average of the values in each cell
> (TT['cat', ] + TT['dog', ] + TT['time', ]) / 3
      breed      tail      feed      kill important explain likely
227.3      13.0      23.0      78.3      31.7      16.0      34.0
```

"time is the cause not the effect"

```
round(context.vectors(TT, s2),3)
      breed tail feed      kill important explain likely
1 6.333 3.333   10 47.667   70.333 38.667    55
```

Context vectors: can we do them in **wordspace**?

Almost there ...

```
# context.vectors() can also take multiple sentences as input
contexts <- context.vectors(TT, c(s1, s2))
# the output is a matrix, let's label the rows "s1" and "s2"
rownames(contexts) <- c("s1", "s2")
# ... and then append it to our original matrix
TT <- rbind(TT, contexts)
TT
      breed tail feed      kill important explain likely
cat      84.00 17.00   8 38.00         0.00   2.00     0
dog     579.00 14.00  32 63.00         1.00   2.00     2
animal   45.00 11.00  86 136.00        13.00   5.00     4
time     19.00  8.00  29 134.00        94.00  44.00    100
reason    1.00  0.00   1 18.00        71.00 140.00     39
cause     0.00  1.00   0   3.00        55.00  35.00     51
effect    0.00  1.00   1   6.00        62.00  37.00     14
s1      227.33 13.00  23 78.33        31.67  16.00     34
s2       6.33  3.33  10 47.67        70.33  38.67     55
```

Context vectors: can we do them in **wordspace**?

And what now?

we can do the same things now as with other DSM matrices

including nearest neighbors ...

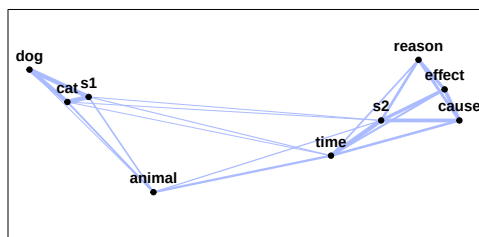
nearest.neighbours(TT, c("s1", "s2"), n=6)

```
$s1
      cat      dog animal      time      s2 cause
14.31016 17.16200 55.27587 62.66470 67.81707 77.90557

$s2
      time cause effect reason animal      s1
18.85097 25.19348 31.51682 40.83768 60.61621 67.81707
```

Context vectors: can we do them in **wordspace**?

```
# ... and a semantic map!
plot(dist.matrix(TT))
```



`hands_on_part4.R` also contains an example for *bank* with `word2vec` embeddings. If you enjoy centroids, try `hands_on_part4_schuetze1998.R`!

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Compositionality

Can we capture it in DS?

- Formally: compositionality implies some operator $\oplus$ such that

$$\text{meaning}(w_1 w_2) = \text{meaning}(w_1) \oplus \text{meaning}(w_2)$$
- CDSM recipe
 - **Distributional vectors** $\mathbf{v}_1$ for w_1 and $\mathbf{v}_2$ for w_2
 - **Operators**: mathematical strategies to combine $\mathbf{v}_1$ and $\mathbf{v}_2$ to predict a vector representation $\mathbf{v}_1 \oplus \mathbf{v}_2$ for $w_1 w_2$
 - ★ vector addition
 - ★ vector multiplication
 - ★ nonlinear operations learned by neural networks
- Problem: some words (e.g. *not*) are themselves more like operators than points in space

Great overview paper: [Frege in space: A program for compositional distributional semantics](#) (Baroni et al. 2014)

Compositionality with distributional vectors

Additive and Multiplicative Models (Mitchell & Lapata 2010)

	music	solution	economy	craft	create
practical	0	6	2	10	4
difficulty	1	8	4	4	0
problem	2	15	7	9	1

$$\mathbf{w} = \mathbf{u} + \mathbf{v}$$

$$\text{predicted}(\text{practical difficulty}) = \mathbf{practical} + \mathbf{difficulty} = [1 \ 14 \ 6 \ 14 \ 4]$$

$$\mathbf{w} = \mathbf{u} \odot \mathbf{v}$$

$$\text{predicted}(\text{practical difficulty}) = \mathbf{practical} \odot \mathbf{difficulty} = [0 \ 48 \ 8 \ 40 \ 0]$$

What is your intuition about the effect of multiplication? How is it different from addition? Have you already seen it as an ingredient of something else?

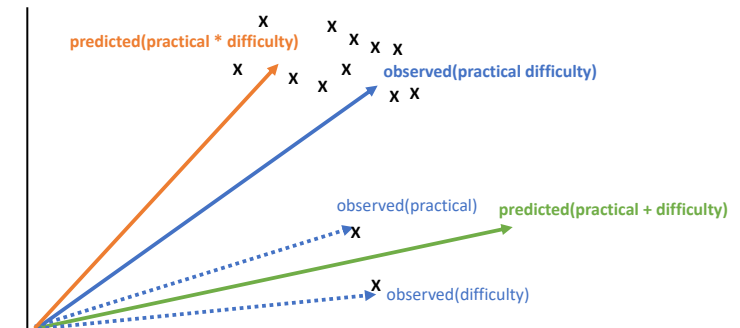
How do I know my compositional vectors are “good”?

Evaluation, again :)

- Qualitative inspection** of nearest neighbors of $\mathbf{u} \oplus \mathbf{v}$
 - In which case look the neighbours intuitively more sensible?
 - problem** is a “good” neighbour of **practical** $\oplus$ **difficulty**
- Quantitative evaluation**
 - Collect DSM vector for *practical difficulty* in the same corpus
 - Evaluation criterion:
observed(practical difficulty) $\approx$ predicted(practical difficulty)
 - ★ Would you expect + or $\odot$ to work better?
 - Evaluation metric
 - ★ distance(predicted, observed) (Lazaridou et al. 2013)
 - ★ neighbour rank (Baroni & Zamparelli 2010; Padó et al. 2016)

How do I know my compositional vectors are “good”?

Observed vs. predicted vector



rank(predicted(practical + difficulty)) = 5 < rank(predicted(practical * difficulty)) = 10

distance(predicted(practical * difficulty)) < distance(predicted(practical + difficulty))

Adjective-noun composition (Baroni & Zamparelli 2010)

Starting point: observed AN vectors

- Input: triples of {observed(AN), A, N}
 - {bad luck, bad, luck}, {red cover, red, cover}, etc.
 - 36 adjectives (size, color, temporal, etc.)

<i>bad luck</i>	<i>electronic communities</i>	<i>historical map</i>
bad	electronic storage	topographical
bad weekend	electronic transmission	atlas
good spirit	purpose	historical material
<i>important route</i>	<i>nice girl</i>	<i>little war</i>
important transport	good girl	great war
important road	big girl	major war
major road	guy	small war
<i>red cover</i>	<i>special collection</i>	<i>young husband</i>
black cover	general collection	small son
hardback	small collection	small daughter
red label	archives	mistress

- Methods: increasing computational complexity
 - unsupervised (additive, multiplicative)
 - supervised: learns matrix A by comparing AN and N

Adjective-noun composition (Baroni & Zamparelli 2010)

Observed(AN) vs. predicted(AN): neighbours

SIMILAR			DISSIMILAR		
<i>adj N</i>	<i>obs. neighbor</i>	<i>pred. neighbor</i>	<i>adj N</i>	<i>obs. neighbor</i>	<i>pred. neighbor</i>
common understanding	common approach	common vision	American affair	Am. development	Am. policy
different authority	diff. objective	diff. description	current dimension	left (a)	current element
different partner	diff. organisation	diff. department	good complaint	current complaint	good beginning
general question	general issue	<i>same</i>	great field	excellent field	gr. distribution
historical introduction	hist. background	<i>same</i>	historical thing	different today	hist. reality
necessary qualification	nec. experience	<i>same</i>	important summer	summer	big holiday
new actor	new cast	<i>same</i>	large pass	historical region	large dimension
recent request	recent enquiry	<i>same</i>	special something	little animal	special thing
small drop	droplet	drop	white profile	chrome (n)	white show
young engineer	young designer	y. engineering	young photo	important song	young image

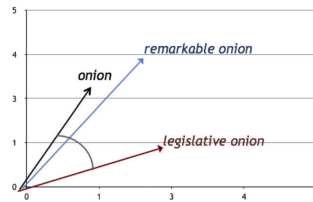
Table 4: Left: nearest neighbors of observed and *alm*-predicted ANs (excluding each other) for a random set of ANs where rank of observed w.r.t. predicted is 1. Right: nearest neighbors of predicted and observed ANs for random set where rank of observed w.r.t. predicted is $\geq 1K$.

How about unattested AN combinations?

Capturing semantically deviant AN combinations (Vecchi *et al.* 2017)

Can we use compositional DSMs to tell, among equally unattested AN, which one is semantically less plausible?

The *composed vectors* for semantically deviant combinations (human ratings) will be **farther away** from the head noun than the acceptable ones



... they test other measures (e.g., neighbors density, vector length) as well as different composition methods: have a look at the paper!

How about unattested AN combinations?

Capturing semantically deviant AN combinations (Vecchi *et al.* 2017)

Can we use compositional DSMs to tell, among equally unattested AN, which one is semantically less plausible?

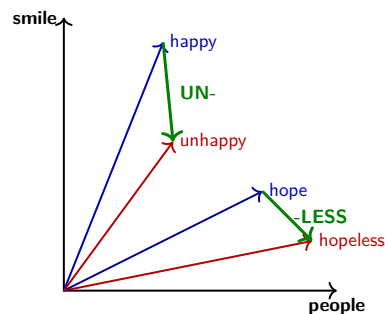
Qualitative inspection: the *composed vectors* of semantically acceptable pairs have plausible nearest neighbors

- a. *angry lamp { *shocked, fearful, angry, defiant* }
- b. *nuclear fox { *nuclear, nuclear arm, nuclear development, nuclear expert* }
- c. warm garlic { *green salad, wild mushroom, sauce, green sauce* }
- d. spectacular striker { *goal, crucial goal, famous goal, amazing goal* }

`hands_on_part4.R` contains an implementation of vector addition and multiplication in wordspace. Have fun chasing the strangest AN combinations (and others as well)!

Compositionality below the word level

Can we use compositional DSMs to investigate the meaning of derivational patterns?



- ▶ Starting point: vectors for **base** and **derived** words.
- ▶ Two strategies:
 - ✎ learn the **semantic shifts** with compositional methods
 - ▶ investigate **properties** of the **patterns** → semantic relations
 - ★ zero-nominalisations as hyponyms of the base verb (Varvara *et al.* 2021)
 - ★ *un-* as antonyms of the base nouns

The DS of Derivational Morphology (Lazaridou *et al.* 2013)

1. Input: derived/stem vector pairs for each affix
 - ▶ *un-*: unfaithful/fairful, unbiased/biased, unwell/well
 - ▶ *-ly*: true/truly, mad/madly, deep/deeply
2. Goal: **build one representation per affix**
 - ▶ No (well, little) learning (additive and multiplicative)
 - ★ *un-* = centroid(unfaithful, unbiased, unwell, etc.)
 - ▶ Increasingly complex learning
 - ★ parameters set during training to optimize composition, affixes as matrices (cf. adjectives)
3. Prediction & evaluation
 - ▶ Apply affix to unseen base and compare predicted(derived) *vs.* observed(derived)
 - ★ simplest (additive) & most complex (lexical functional, theoretically motivated): comparable
 - ★ cf. Padó *et al.* (2016) for German: simple composition methods work better!

The DS of Derivational Morphology (Lazaridou *et al.* 2013)

Dataset

Affix	Stem/Der. POS	Training Items	HQ/Tot. Test Items	Avg. SDR
-able	verb/adj	177	30/50	5.96
-al	noun/adj	245	41/50	5.88
-er	verb/noun	824	33/50	5.51
-ful	noun/adj	53	42/50	6.11
-ic	noun/adj	280	43/50	5.99
-ion	verb/noun	637	38/50	6.22
-ist	noun/noun	244	38/50	6.16
-ity	adj/noun	372	33/50	6.19
-ize	noun/verb	105	40/50	5.96
-less	noun/adj	122	35/50	3.72
-ly	adj/adv	1847	20/50	6.33
-ment	verb/noun	165	38/50	6.06
-ness	adj/noun	602	33/50	6.29
-ous	noun/adj	157	35/50	5.94
-y	noun/adj	404	27/50	5.25
in-	adj/adj	101	34/50	3.39
re-	verb/verb	86	27/50	5.28
un-	adj/adj	128	36/50	3.23
<i>tot</i>	<i>*/*</i>	6549	623/900	5.52

7000 base/derived pairs from CELEX, 18 patterns, training vs. test (further annotated for base/derived relatedness and vector quality)

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Not all semantic knowledge is distributional

Proper names “answer the purpose of **showing** what thing it is that we are talking about but not of telling anything about it” (Mill, 1843)

- ▶ Intuition: instances of categories such as PER, ORG, etc.
- ▶ Herbelot (2015), standard DSMs: category → instance
 - ▶ “... upon encountering the name *Mr Darcy* for the first time in the novel, a reader will attribute it the representation of the concept MAN and subsequently **specialise** it as per the linguistic contexts in which the name appears”
- ▶ Westera *et al.* (2021), embeddings: instance → category

Function words: some pointers

- ▶ Baroni *et al.* (2012) on quantifiers/entailment, Bernardi *et al.* (2013) on determiners, Padó & Hole (2022) on the polysemy of the German reflexive *sich*

Making sense: collocational profiles vs. DSM vectors

Some exciting open questions for your consideration

- ▶ For an unreduced term-term DSM, each row vector $\mathbf{m}_t$ corresponds to the collocational profile of a target term t
 - ▶ e.g. dependency-structured DSM → word sketches
 - ▶ compare our reading of sketch differences with distributional similarity for nearest neighbours, esp. surprising ones
- ▶ Can dim.-reduced vectors be read as collocational profiles?
 - ▶ map to original features → dense “virtual” collocational profile
 - ▶ compare top- k entries with original sparse profile
- ▶ Do DSM distances use the same information as the human interpretation of collocational profiles?
 - ▶ humans likely to focus on few salient collocates, while DSM might exploit weak correlations across many dimensions
 - ▶ cf. insights from authorship attribution (Evert *et al.* 2017)

Making sense: collocational profiles vs. DSM vectors

Some exciting open questions for your consideration

- ▶ Can we figure out which features account for semantic differences?
 - ▶ normalise vectors: $\|\mathbf{u}\|_2 = \|\mathbf{v}\|_2 = 1$
 - ▶ $(u_i - v_i)^2$ = contribution to (squared) Euclidean distance
 - ▶ $u_i v_i$ = contribution to cosine similarity
- ▶ Collocations = **first-order** association = **syntagmatic**
- ▶ DSM similarity = **second-order** association = **paradigmatic**
 - ▶ widely-held belief in distributional semantics, but is it true?
 - ▶ some applications actually require both first-order and second-order association (e.g. **free association norms**)
 - ▶ we will obtain a few mathematical insights in Part 5

See `hands_on_part4.R` for some example code on how to interpret DSM vectors in terms of collocational profiles.

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Cognitive modelling with DSMs

- ▶ **Why?** – Because we want to know whether DS captures the mental lexical knowledge of human speakers!
- ▶ Task: DSM predicts reaction times in **priming experiments** (Hare *et al.* 2009; Lapesa & Evert 2013)
 - ▶ often just experimental items used for multiple-choice task (e.g. Padó & Lapata 2007; Herdağdelen *et al.* 2009)
 - ▶ cf. tasks constructed from Lazaridou2013 in previous section
 - ▶ data sets of experimental items: **GEK_Items**, **SPP_Items**
- ▶ Task: DSM predicts **EEG potentials** (Murphy *et al.* 2009) or **fMRI brain activation** levels (Mitchell *et al.* 2008)
 - ▶ huge datasets, but tiny and selective vocabulary
- ▶ Task: DSM predicts human **free associations**
 - ▶ often considered a “window into the mental lexicon”
 - ▶ free association norms available for thousands of cue words

Free associations

... a cue into the organization of the mental lexicon?

Which words come to your mind if you hear ...

- ▶ whisky → gin, drink, scotch, bottle, soda
- ▶ giraffe → neck, animal, zoo, long, tall

- ▶ Hypotheses concerning the nature of the underlying process:
 - ▶ Result of learning by contiguity (James 1890) **syntagmatic (1st-order)**
 - ▶ Result of symbolic processes which make use of complex semantic structures (Clark 1970) **paradigmatic (2nd-order)**
- ▶ Large collections available
 - ▶ Edinburgh Associative Thesaurus (**EAT**)
8210 stimuli, 100 subjects (Kiss *et al.* 1973)
 - ▶ University of South Florida Free Association Norms (**USF**)
5019 stimuli, 6000 subjects (Nelson *et al.* 2004)

Syntagmatic vs. paradigmatic relations

Definitions and general assumptions

- ▶ **Syntagmatic** $\longleftrightarrow$ contiguity
 - ▶ Examples: {dog, barks}, {dog, bone}
 - ▶ Words appear together: 1st-order co-occurrence
 - ▶ Found in: collocational profiles, DSM dimensions
- ▶ **Paradigmatic** $\longleftrightarrow$ interchangeability
 - ▶ Examples: {book, volume}, {dog, animal}
 - ▶ Words appear in similar contexts: 2nd-order co-occurrence
 - ▶ Usually semantically related
 - ▶ Found in: DSM nearest neighbours

However ...

DSM neighbourhoods include syntagmatically related words (collocates) if certain parameters are properly set, in particular if the context window is large enough (Lapesa *et al.* 2014).

Free associations in a DSM

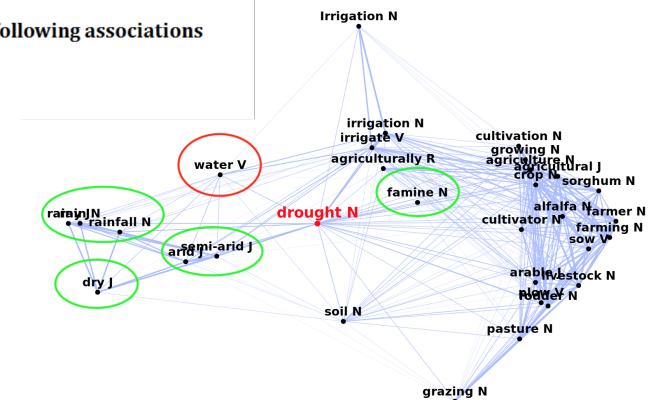
Drought in EAT vs. DSM

drought stimulated the following associations

Number of different answers: 30

Total count of all answers: 97

- WATER 21 0.22
- DRY 16 0.16
- THIRST 9 0.09
- FAMINE 7 0.07
- RAIN 7 0.07
- DESERT 6 0.06
- BEER 5 0.05
- CRACK 2 0.02
- HOT 2 0.02
- SAND 2 0.02
- ALE 1 0.01
- ARID 1 0.01
- AUSTRALIA 1 0.01
- CATTLE 1 0.01
- COLD 1 0.01
- COOL 1 0.01
- DEATH 1 0.01
- DUST 1 0.01
- GALE 1 0.01
- MONSOON 1 0.01



Free associations & co-occurrence data

Previous work

- ▶ Wettler *et al.* (2005)
 - ▶ Data: subset of EAT (100 stimuli)
 - ▶ Task: prediction of the most common free associate
 - ▶ Model: first-order model, BNC, large window (20 words)
 - ▶ Result: human associative responses can be predicted from contiguities between words in language use (collocations)
- ▶ ESSLLI 2008 Shared Task
 - ▶ Data: subset of EAT (a different set of 100 stimuli)
 - ▶ Task 1: discrimination btw. the most common associate and hapax/random distractors → multiple choice
 - ▶ Task 2: prediction of the most common free associate
 - ▶ Result: first-order models (collocations) are better than second-order models (DSMs)

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Compositionality
Other linguistic aspects

The FAST task

Free association norms
FAST: Data set and tasks
FAST: Experiments
Hands-on exercises

The Free ASsociation Task (FAST) data set

Preprocessing

1. Starting point: EAT (8210 stimuli), USF (5019 stimuli)
2. Out-of-context POS tagging
 - ▶ Annotate items in EAT and USF (stimuli and responses) with part of speech information
 - ▶ How? Most frequent POS in Web corpus ENCOW: publicly available 10-billion-word Web corpus → replicability
3. Out-of-context lemmatization
 - ▶ **morpha**, a robust morphological analyzer
<http://users.sussex.ac.uk/~johnca/morph.html>
 - ▶ lemmatization of unknown words based on POS tag
4. Annotation with frequency information
 - ▶ frequency lists from ENCOW (lemmatised with **morpha**)

The Free ASsociation Task (FAST) data set

Item selection

For each stimulus in EAT (8210) and USF (5019) select a:

- ▶ **FIRST**: the most common associate response
- ▶ **HAPAX**: a response generated for the target once
 - ▶ or twice for USF (hapax responses are omitted there)
 - ▶ if several HAPAX candidates are available, pick the one whose lemma frequency matches most closely that of **FIRST**
- ▶ **RANDOM**, by randomly picking a word which was among the top 25% associates of *another stimulus* (and produced at least 5 times). If possible:
 - ▶ match lemma frequency of **RANDOM** and **FIRST**
 - ▶ try to use each **RANDOM** only once

(multiwords, numbers, closed-class words, and other words that do not occur in ENCOW were discarded)

The FAST data set

Final data set

- ▶ EAT subset: **3836** test items + **3774** training items
- ▶ USF subset: **2359** test items + **2360** training items
- ▶ Item = (STIMULUS, FIRST, HAPAX, RANDOM)
- ▶ Each stimulus and candidate response provided as lowercased word form and POS-disambiguated lemma
 - + ENCOW frequency information
 - + # test subjects who produced response
- ▶ Download: <https://osf.io/cd8ar/> (Evert & Lapesa 2021)
- ▶ Included as **FAST** in package **wordspaceEval**

The FAST dataset

The new EAT task isn't perfect either ... yet

- ▶ Guessing POS from corpus doesn't always work
 - ▶ e.g. *fit*_{VERB} → *epileptic*_{ADJ}, *aristocracy*_{NOUN} → *lords*_{NAME}
 - ▶ but very few lemmatization errors (e.g. *daiquiri* → *daiquirus*)
- ▶ Colloquialisms and British slang
 - ▶ e.g. *bod*_{NOUN} → *person*_{NOUN} (rare in written corpus)
 - ▶ but Web corpus has Welsh *bod* 'to be' mistagged as noun
 - ▶ DSM neighbours: *yn*, *hynny*, *mewn*, *hwn*, *gyfer*, ..., 49. *bloke*, *techy*_{NOUN}, *nus*, *hon*, ..., 60. *guy*, *mai*, *geezer*, ...
 - ▶ another example is *mellow*_{ADJ} → *yellow*_{ADJ}

The FAST tasks

Task 1: **multiple-choice**

- ▶ Given a stimulus and a $\langle \text{FIRST}, \text{HAPAX}, \text{RANDOM} \rangle$ triple, determine which of the three candidates is FIRST.
 - ▶ Stimulus: *accept*, $\langle \text{receive}, \text{love}, \text{soul} \rangle$
- ▶ **Performance:** accuracy
- ▶ **Baseline:** 33.3%

The FAST tasks

Task 2: **open-vocabulary lexical access**

- ▶ Given a stimulus (e.g., *accept*), predict FIRST (*receive*) out of a candidate set (all FIRST: USF=1197, EAT=1633)
- ▶ **Performance:** two measures
 - ▶ **Soft accuracy:** average over reciprocal rank ($1/r$) of the true FIRST associate, as a percentage.
 - ★ similar to accuracy of predicting first associate, but awards partial points for almost correct guesses
 - ★ always $\geq$ top-1 accuracy
 - ▶ **Log rank:** geometric mean of r across all stimuli.
 - ★ corresponds to average over $\log r$
 - ★ better differentiation for models that rarely get the correct answer (and hence score low on soft accuracy)
- ▶ **Baselines**
 - ▶ **Soft accuracy:** USF=0.64% and EAT=0.49%
 - ▶ **Log rank:** USF=442.0 and EAT=602.4

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FAST: Data set and tasks

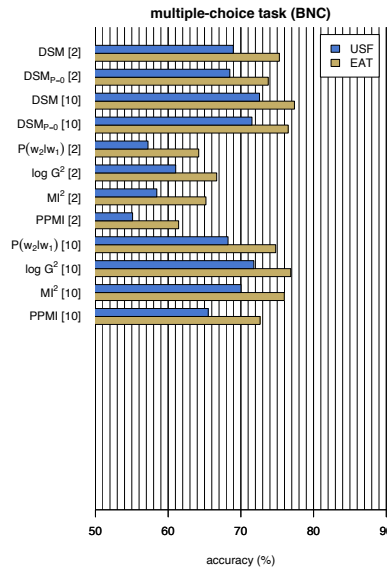
FAST: Experiments

Hands-on exercises

Experimental setup

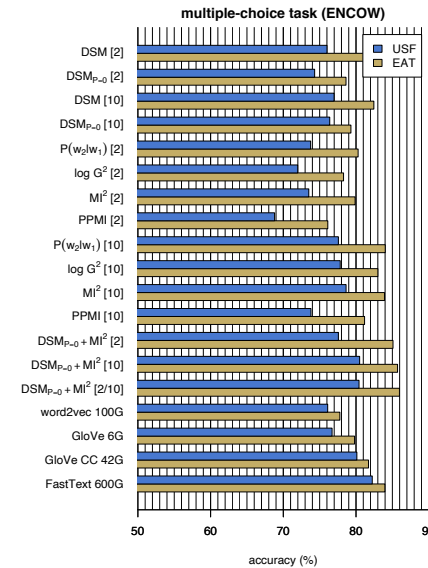
- ▶ **DSMs (second-order):** symmetric span of 2 *vs.* 10 words, other parameters set according to Lapesa & Evert (2014).
 - ▶ we experiment with Caron P (Bullinaria & Levy 2012)
 - ▶ $P = 0$ equalizes contributions of SVD dimensions
- ▶ **Collocations (first-order):** symmetric span, 2 *vs.* 10 words, with four different association measures (Evert 2008)
 - ▶ conditional probability $P(w_2|w_1)$
 - ▶ log-likelihood $\log G^2$ (popular for collocations)
 - ▶ $MI^2 = \log_2 \frac{O^2}{E} =$ geometric mean of $P(w_2|w_1)$ and $P(w_1|w_2)$
 - ▶ PPMI (popular for DSMs)
- ▶ **Corpus data:** for DSMs and collocations
 - ▶ British National Corpus: 100M words
 - ▶ ENCOW 2014 Web corpus, unique sentences: 8.5G words
- ▶ **Neural embeddings:** pre-trained models
 - ▶ word2vec (Mikolov *et al.* 2013): 100G tokens of Google News
 - ▶ GloVe (Pennington *et al.* 2014): 6G tokens Wikipedia + Gigaword
 - ▶ GloVe: 42G tokens Web data (Common Crawl)
 - ▶ FastText (Joulin *et al.* 2017): 600G tokens Common Crawl

Results: Multiple-choice task



- ▶ British National Corpus (100M words)
- ▶ EAT vs. USF

Results: Multiple-choice task



- ▶ ENCOW 2014 Web (8.5G words)
- ▶ EAT vs. USF
- ▶ Embeddings trained on much larger corpora
- ▶ Combined 1st-/2nd-order
 - ▶ DSM_{P=0} + MI²
 - ▶ using neighbour rank
 - ▶ harmonic mean
 - ▶ competitive with state-of-the-art embeddings

Results: Multiple-choice task

model	span	$n = 2359$	$n = 3836$
		USF	EAT
DSM	2	76.01%	81.78%
DSM _{P=0}	2	74.31%	78.62%
DSM	10	76.98%	82.46%
DSM _{P=0}	10	76.39%	79.30%
$P(w_2 w_1)$	10	77.58%	84.02%
$\log G^2$	10	77.83%	83.00%
MI ²	2	78.64%	83.92%
PPMI	10	73.80%	81.18%
Combined	2	77.58%	85.09%
Combined	10	80.50%	85.71%
Combined mix		80.41%	85.97%
word2vec	–	76.11%	77.78%
GloVe	–	76.71%	79.80%
GloVe CC	–	80.12%	81.72%
FastText	–	82.24%	83.97%

- ▶ ENCOW 2014 Web (8.5G words)
- ▶ EAT vs. USF
- ▶ Embeddings trained on much larger corpora
- ▶ Combined 1st-/2nd-order
 - ▶ DSM_{P=0} + MI²
 - ▶ using neighbour rank
 - ▶ harmonic mean
 - ▶ competitive with state-of-the-art embeddings

Results: Open-choice task

model	span	$n = 2359$		$n = 3836$	
		USF		EAT	
		soft acc.	lrank	soft acc.	lrank
DSM	2	41.54%	6.6	34.53%	9.9
DSM _{P=0}	2	42.12%	7.6	34.67%	12.1
DSM	10	42.01%	6.0	35.93%	9.1
DSM _{P=0}	10	42.86%	7.1	35.68%	11.6
$P(w_2 w_1)$	10	22.34%	17.0	11.27%	27.1
$\log G^2$	10	37.63%	6.6	34.13%	8.8
MI ²	10	39.73%	6.2	34.01%	8.7
PPMI	10	35.34%	8.2	29.29%	12.2
Combined	2	42.29%	5.5	37.54%	7.0
Combined	10	44.99%	4.8	39.48%	6.5
Combined mix		45.36%	4.8	39.48%	6.4
word2vec	–	38.98%	7.7	30.51%	14.8
GloVe	–	39.22%	7.6	30.19%	13.8
GloVe CC	–	44.01%	5.7	34.26%	10.5
FastText	–	51.00%	4.1	40.34%	7.2

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Bonus task: Reverse free associations

The CogALex-IV shared task (Rapp & Zock 2014)

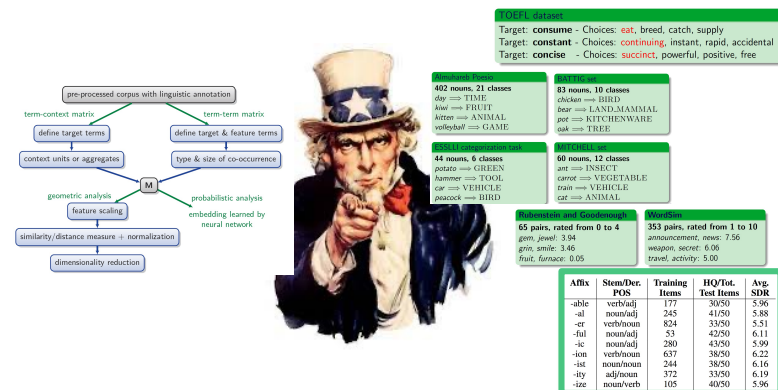
Reverse multiword free association

- ▶ wheel, driver, bus, drive, lorry → ?
- ▶ away, minded, gone, present, ill → ?
- ▶ Data: subset of EAT (2000 stimuli each training/test)
- ▶ Very challenging (best: 35% accuracy)
 - ▶ open-ended vocabulary (including inflected surface forms!)
 - ▶ need for integrating predictions of different stimuli
- ▶ And the winner was ...
 - ▶ a system using first-order statistics to re-rank the output of a “standard” DSM (Ghosh *et al.* 2015)
 - ▶ our submission: best 1st-order: 27.7% / best 2nd-order: 14.0%
- ▶ Try it yourself: `CogALex4.rda` in `wordspaceEval`

Hands-on exercise

- ▶ Solve the FAST multiple-choice task with a DSM
 - ▶ `eval.multiple.choice()` does most of the work for you
 - ▶ use `details=TRUE` to inspect biggest mistakes and explore performance (e.g. wrt. frequency of stimulus and response)
- ▶ Can you also make use of first-order (collocation) data?
 - ▶ hint: the DSM matrix **M** contains co-occurrence counts
- ▶ Advanced: Can you combine DSMs with first-order data?
 - ▶ hint: use average of DSM and first-order “neighbour” rank
- ▶ Advanced: Try to solve the open-choice lexical access task
 - ▶ no ready-made evaluation function in `wordspace` yet
- ▶ R code in `hands_on_part4_fast.R` will help you get started!

It is practice session time!



References I

- Baroni, Marco and Zamparelli, Roberto (2010). Nouns are vectors, adjectives are matrices: Representing adjective-noun constructions in semantic space. In *Proceedings of the 2010 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1183–1193, MIT, Massachusetts, USA.
- Baroni, Marco; Bernardi, Raffaella; Do, Ngoc-Quynh; Shan, Chung-chieh (2012). Entailment above the word level in distributional semantics. In *Proceedings of the 13th Conference of the European Chapter of the Association for Computational Linguistics (EACL 2012)*, pages 23–32, Avignon, France.
- Baroni, Marco; Bernardi, Raffaella; Zamparelli, Roberto (2014). Frege in space: A program for compositional distributional semantics. *Linguistic Issues in Language Technology (LiLT)*, 9(6), 5–109.
- Bernardi, Raffaella; Dinu, Georgiana; Marelli, Marco; Baroni, Marco (2013). A relatedness benchmark to test the role of determiners in compositional distributional semantics. In *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics (Volume 2: Short Papers)*, pages 53–57, Sofia, Bulgaria. Association for Computational Linguistics.
- Boleda, Gemma (2020). Distributional semantics and linguistic theory. *Annual Review of Linguistics*, 6(1), 213–234.

References III

- Evert, Stefan; Proisl, Thomas; Jannidis, Fotis; Reger, Isabella; Pielström, Steffen; Schöch, Christof; Vitt, Thorsten (2017). Understanding and explaining Delta measures for authorship attribution. *Digital Scholarship in the Humanities*, 22(suppl_2), ii4–ii16.
- Ghosh, Urmi; Jain, Sambhav; Paul, Soma (2015). A two-stage approach for computing associative responses to a set of stimulus words. In Z. (eds.) (ed.), *Proceedings of the 4th Workshop on Cognitive Aspects of the Lexicon*.
- Hare, Mary; Jones, Michael; Thomson, Caroline; Kelly, Sarah; McRae, Ken (2009). Activating event knowledge. *Cognition*, 111(2), 151–167.
- Herbelot, Aurélie (2015). Mr Darcy and Mr Toad, gentlemen: distributional names and their kinds. In *Proceedings of the 11th International Conference on Computational Semantics*, pages 151–161, London, UK. Association for Computational Linguistics.
- Herdağdelen, Amaç; Erk, Katrin; Baroni, Marco (2009). Measuring semantic relatedness with vector space models and random walks. In *Proceedings of the 2009 Workshop on Graph-based Methods for Natural Language Processing (TextGraphs-4)*, pages 50–53, Suntec, Singapore.
- James, W (1890). *The Principles of Psychology*. Dover, New York.

References II

- Bullinaria, John A. and Levy, Joseph P. (2012). Extracting semantic representations from word co-occurrence statistics: Stop-lists, stemming and SVD. *Behavior Research Methods*, 44(3), 890–907.
- Clark, H.H. (1970). Word associations and linguistic theory. In J. Lyons (ed.), *New horizons in linguistics*. Harmondsworth: Penguin.
- Curran, James; Clark, Stephen; Bos, Johan (2007). Linguistically motivated large-scale NLP with C&C and Boxer. In *Proceedings of the 45th Annual Meeting of the Association for Computational Linguistics, Posters and Demonstrations Sessions*, pages 33–36, Prague, Czech Republic.
- Evert, Stefan (2008). Corpora and collocations. In A. Lüdeling and M. Kytö (eds.), *Corpus Linguistics. An International Handbook*, chapter 58, pages 1212–1248. Mouton de Gruyter, Berlin, New York.
- Evert, Stefan (2016). CogALex-V shared task: Mach5 – a traditional DSM approach to semantic relatedness. In *Proceedings of the 5th Workshop on Cognitive Aspects of the Lexicon (CogALex-V)*, pages 92–97, Osaka, Japan.
- Evert, Stefan and Lapasa, Gabriella (2021). FAST: A carefully sampled and cognitively motivated dataset for distributional semantic evaluation. In *Proceedings of the 25th Conference on Computational Natural Language Learning (CoNLL 2021)*, pages 588–595, Online. Data set: <https://osf.io/cd8ar/>.

References IV

- Joulin, Armand; Grave, Edouard; Bojanowski, Piotr; Mikolov, Tomas (2017). Bag of tricks for efficient text classification. In *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers*, pages 427–431, Valencia, Spain.
- Kiss, G. R.; Armstrong, C.; Milroy, R.; Piper, J. (1973). An associative thesaurus of English and its computer analysis. In A. Aitken, R. Beiley, and N. Hamilton-Smith (eds.), *The Computer and Literary Studies*. Edinburgh University Press, Edinburgh.
- Lapasa, Gabriella and Evert, Stefan (2013). Evaluating neighbor rank and distance measures as predictors of semantic priming. In *Proceedings of the ACL Workshop on Cognitive Modeling and Computational Linguistics (CMCL 2013)*, pages 66–74, Sofia, Bulgaria.
- Lapasa, Gabriella and Evert, Stefan (2014). A large scale evaluation of distributional semantic models: Parameters, interactions and model selection. *Transactions of the Association for Computational Linguistics*, 2, 531–545.
- Lapasa, Gabriella; Evert, Stefan; Schulte im Walde, Sabine (2014). Contrasting syntagmatic and paradigmatic relations: Insights from distributional semantic models. In *Proceedings of the Third Joint Conference on Lexical and Computational Semantics (*SEM 2014)*, pages 160–170, Dublin, Ireland.

References V

- Lazaridou, Angeliki; Marelli, Marco; Zamparelli, Roberto; Baroni, Marco (2013). Compositionally derived representations of morphologically complex words in distributional semantics. In *Proceedings of the 51st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1517–1526, Sofia, Bulgaria.
- Mikolov, Tomas; Chen, Kai; Corrado, Greg; Dean, Jeffrey (2013). Efficient estimation of word representations in vector space. In *Workshop Proceedings of the International Conference on Learning Representations 2013*.
- Mitchell, Jeff and Lapata, Mirella (2010). Composition in distributional models of semantics. *Cognitive Science*, **34**, 1388–1429.
- Mitchell, Tom M.; Shinkareva, Svetlana V.; Carlson, Andrew; Chang, Kai-Min; Malave, Vicente L.; Mason, Robert A.; Just, Marcel Adam (2008). Predicting human brain activity associated with the meanings of nouns. *Science*, **320**, 1191–1195.
- Murphy, Brian; Baroni, Marco; Poesio, Massimo (2009). EEG responds to conceptual stimuli and corpus semantics. In *Proceedings of the 2009 Conference on Empirical Methods in Natural Language Processing*, pages 619–627, Singapore.
- Nelson, Douglas L.; McEvoy, Cathy L.; Schreiber, Thomas A. (2004). The University of South Florida word association, rhyme, and word fragment norms. *Behavior Research Methods, Instruments, & Computers*, **36**(3), 402–407.

References VI

- Padó, Sebastian and Lapata, Mirella (2007). Dependency-based construction of semantic space models. *Computational Linguistics*, **33**(2), 161–199.
- Padó, Sebastian; Herbelot, Aurélie; Kisselew, Max; Šnajder, Jan (2016). Predictability of distributional semantics in derivational word formation. In *Proceedings of COLING 2016, the 26th International Conference on Computational Linguistics: Technical Papers*, pages 1285–1296, Osaka, Japan. The COLING 2016 Organizing Committee.
- Padó, Sebastian and Hole, Daniel (2022). Distributional analysis of polysemous function words. In *Proceedings of the 13th International Tbilisi Symposium on Language, Logic and Computation (TbiLLC 2019)*, volume 13206 of *Lecture Notes in Computer Science*. Springer.
- Pennington, Jeffrey; Socher, Richard; Manning, Christopher D. (2014). GloVe: Global vectors for word representation. In *Proceedings of EMNLP 2014*.
- Rapp, Reinhard and Zock, Michael (2014). The cogalex-iv shared task on the lexical access problem. In *Proceedings of the 4th Workshop on Cognitive Aspects of the Lexicon*, pages 1–14. Zock/Rapp/Huang (eds.).

References VII

- Santus, Enrico; Gladkova, Anna; Evert, Stefan; Lenci, Alessandro (2016). The CogALex-V shared task on the corpus-based identification of semantic relations. In *Proceedings of the 5th Workshop on Cognitive Aspects of the Lexicon (CogALex-V)*, pages 69–79, Osaka, Japan.
- Schäfer, Roland and Bildhauer, Felix (2012). Building large corpora from the web using a new efficient tool chain. In *Proceedings of the Eighth International Conference on Language Resources and Evaluation (LREC '12)*, pages 486–493, Istanbul, Turkey. ELRA.
- Schütze, Hinrich (1998). Automatic word sense discrimination. *Computational Linguistics*, **24**(1), 97–123.
- Varvara, Rossella; Lapesa, Gabriella; Padó, Sebastian (2021). Grounding semantic transparency in context: A distributional semantic study on German event nominalizations. *Morphology*.
- Vecchi, Eva M.; Marelli, Marco; Zamparelli, Roberto; Baroni, Marco (2017). Spicy adjectives and nominal donkeys: Capturing semantic deviance using compositionality in distributional spaces. *Cognitive Science*, **41**(1), 102–136.
- Westera, Matthijs; Gupta, Abhijeet; Boleda, Gemma; Padó, Sebastian (2021). Distributional models of category concepts based on names of category members. *Cognitive Science*, **45**(e13029).

References VIII

- Wettler, Manfred; Rapp, Reinhard; Sedlmeier, Peter (2005). Free word associations correspond to contiguities between words in texts*. *Journal of Quantitative Linguistics*, **12**(2–3), 111–122.